Tim Kam Lun TSANG

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Research Focus: Biostatistics & Health Data Science
Integrating statistical modelling, causal inference, and AI to forecast infectious disease dynamics (COVID-19, Influenza) and evaluate intervention policies.
Keywords: Population Immunity, Real-time Surveillance, Transmission Dynamics, Public Health Data Science.

Education

- Postdoctoral Associate**,** Department of Biostatistics, College of Public Health and Health Professions, University of Florida, Oct 2016 – Jun 2019
- Ph.D., Biostatistics and Epidemiology, World Health Organization (WHO) Collaborating Centre for Infectious Disease Epidemiology and Control, School of Public Health, The University of Hong Kong, May 2016
- M.Phil., Mathematics and Statistics, Department of Mathematics, Hong Kong University of Science and Technology, Aug 2012
- B.Sc. Mathematics and Statistics**,** Department of Mathematics, Hong Kong University of Science and Technology, May 2010

Employment

- Assistant Professor, School of Public Health, Li Ka Shing Faculty of Medicine, The University of Hong Kong, Oct 2021 - present
- Research Assistant Professor, School of Public Health, Li Ka Shing Faculty of Medicine, The University of Hong Kong, Jul 2019 – Sep 2021
- Senior Technical Officer**,** WHO Collaborating Centre for Infectious Disease Epidemiology and Control, School of Public Health, The University of Hong Kong, May - Sep 2016

Grant Awards

Ongoing Grants:

1. HKUMed Research Collaboration Booster Fund 2025 2026-2028
   Comparative Analysis of Household Transmission Dynamics and Intervention Strategies for Influenza and COVID-19
   Role: PI. Funding amount: 1,000,000
2. URC PDF/RAP Scheme (49th Round) 2026-2028
   Methodologies for Estimating Population Immunity Against Infectious Diseases
   Role: PI. Funding amount: 50% Post-doctoral Fellow for 3 years
3. RGC General Research Fund (GRF) (Ref. 17106424) 2025-2026
   Impact of school-based control measure on population transmission dynamics of COVID-19 in Hong Kong
   Role: PI. Funding amount: 830,240
4. HKUMed Enhanced New Staff Start-up Research Grant 2023-2026
   Risk factors of transmission and variation of individual infectiousness of COVID-19
   Role: PI. Funding amount: 380,000

5. Health and Medical Research Fund (HMRF) (Ref. 23242891) (PI: Yang) 2026-2028
Bridging potential and realized healthcare utilization: A comprehensive population-level analysis of financial, geographic, and literacy barriers in Hong Kong
Role: Co-PI. Funding amount: 1,150,000
6. RGC GRF (Ref. 17117525) (PI: Dhanasekaran) 2026-2028
Estimating epidemiological parameters from viral outbreak phylogenies
Role: Co-I. Funding amount: 1,331,776
7. Health and Medical Research Fund (HMRF) (Ref. 24230942) (PI: Yang) 2025-2027
Evaluating the influence of childhood immune imprinting on birth cohort-specific disease burden of seasonal influenza A in Hong Kong
Role: Co-PI. Funding amount: 775,788
8. RGC GRF (Ref. 17111124) (PI: Ali) 2025-2026
Inferring seasonality and population immunity for projecting future dengue dynamics and associated burden in Asian countries using multi-stream spatiotemporal data
Role: Co-I. Funding amount: 1,146,291

Completed Grants:

9. HMRF (Ref. 21200292) 2022-2025
Employing school absenteeism data for nowcasting and forecasting of influenza epidemics in Hong Kong
Role: PI. Funding amount: 851,680
10. RGC GRF (Ref. 17100822) 2023-2024
Heterogeneity of individual infectiousness and its implications on population transmission dynamics of COVID-19 in Hong Kong
Role: PI. Funding amount: 713,338
11. RGC GRF (Ref. 17110221) 2022-2023
12. Impact of unidentified cases on monitoring real-time transmissibility and effectiveness of control measures of COVID-19 in Hong Kong
Role: PI. Funding amount: 796,467
13. HMRF Commissioned Research (Ref. CID-HKU2-10) 2022-2023
Monitoring real-time transmissibility and potential of superspreading of COVID-19 in Hong Kong
Role: PI. Funding amount: 940,166
14. HMRF (Ref: 20190542) 2021-2023
How infection history shapes individual and population immunity against influenza in Hong Kong
Role: PI. Funding amount: 916,992
15. HMRF, Research Fellowship Scheme (Ref: 05190097) 2020-2022
Identifying optimal age groups for influenza vaccination as a transmission-limiting strategy to reduce the impact of influenza in Hong Kong
Role: PI. Funding amount: 889,000
16. RGC GRF (Ref: 17104220) 2020-2022
Estimating the infectiousness of asymptomatic influenza virus infections and the consequences for public health control measures
Role: PI. Funding amount: 1,122,805
17. Research Assessment Exercise (RAE) 2026 Preparation, HKU 2025
Nowcasting and forecasting time-varying effective reproduction number of COVID-19 in Hong Kong
Role: PI. Funding amount: 100,000
18. Seed Fund for Basic Research, HKU 2025-2026

Exploring pre-symptomatic transmissions potential of influenza in Hong Kong
Role: PI. Funding amount: 88,900

19. Seed Fund for Basic Research, HKU 2024
Nowcasting and Forecasting time-varying effective reproductive number of COVID-19 and influenza with incorporating temporal predictors in Hong Kong
Role: PI. Funding amount: 88,900

20. Seed Fund for Basic Research, HKU 2023
Incorporating temporal predictors for forecasting of COVID-19 case numbers in Hong Kong
Role: PI. Funding amount: 150,000

21. Seed Fund for Basic Research for new staff, HKU 2022
Quantifying transmission heterogeneity of COVID-19 in households
Role: PI. Funding amount: 143,500

22. RGC, Senior Research Fellowship (Ref: SRFS2021-7S03) (PI: Cowling) 2021-2025
Determining optimal influenza vaccination strategies in Hong Kong
Role: Co-I. Funding amount: 7,798,380

23. RGC, Collaborative Research Fund (Ref: C7123-20G) (PI: Cowling) 2021-2024
Superspreading of COVID-19: epidemiology and control
Role: Co-I. Funding amount: 8,978,751

24. HMRF (Ref. 22210552) (PI: Yang) 2023-2025
Incorporating population viral loads to real-time estimation of the transmission of SARS-CoV-2 variants among population with pre-existing immunity
Role: Co-I. Funding amount: 869,760

25. HMRF (Ref. 22211192) (PI: Adam) 2023-2025
Time-varying infectivity, susceptibility and risk factors associated with SARS-CoV-2 superspreading dynamics
Role: Co-I. Funding amount: 500,000

26. HMRF (Ref. 20190502) (PI: Leung) 2021-2025
Transmissibility of respiratory viruses
Role: Co-I. Funding amount: 1,500,000

27. HMRF (Ref. 21200212) (PI: Yang) 2022-2024
Optimizing border control measures to prevent importations of COVID-19 or another pandemic infection
Role: Co-I. Funding amount: 945,044

28. HMRF (Ref. 21200352) (PI: Ali) 2022-2024
Assessment of seasonality of influenza and COVID-19 transmission to predict their future dynamics and burden in Hong Kong
Role: Co-I. Funding amount: 1,021,680

29. HMRF (Ref. 21200632) (PI: Du) 2022-2024
Proactive COVID-19 testing strategies in the time of mass vaccination in Hong Kong
Role: Co-I. Funding amount: 1,021,680

Publications

Selected 5 Key Publications

1. Xiong W, Huang X, Tran TTN, Nguyen TLT, Peng L, Perera RAPM, Fang VJ, Yang B, Wong SS, Leung NHL, Sullivan SG, Ip DKM, Peiris M, Ross TM, Boni MF, Cowling BJ, **Tsang TK#.** Measuring Population Immunity Against Influenza Using Individual Antibody Titers: An Retrospective Observational Study. ***Accepted by Lancet Infectious Disease***
[IF 56.3 (2022), rank 1/96 infectious disease]

2. Zhang C, Fang VJ, Chan KH, Leung GM, Ip DKM, Peiris JSM, Cowling BJ, Cauchemez S, **Tsang TK#.** Frequent presymptomatic household transmission of influenza A but not influenza B virus. ***Nature Health*** (2026).
3. **Tsang TK#**, Du Q, Cowling BJ, Viboud C. An adaptive weight ensemble approach to forecast influenza activity in an irregular seasonality context. ***Nat Commun***. 2024 Oct 4;15(1):8625.
   [IF 16.6 (2022), rank 6/73 Multidisciplinary Sciences]
4. **Tsang TK#,** Wang C, Fang VJ, Perera RAPM, So HC, Ip DKM, Leung GM, Peiris JSM, Cauchemez S, Cowling BJ. Reconstructing household transmission dynamics to estimate the infectiousness of asymptomatic influenza virus infections. ***Proc Natl Acad Sci U S A.*** 2023 Aug 15;120(33):e2304750120.
   [IF 11.1 (2022), rank 8/73 Multidisciplinary Sciences]
5. **Tsang TK**, Perera RAPM, Fang VJ, Wong JY, Shiu EY, So HC, Ip DKM, Malik Peiris JS, Leung GM, Cowling BJ, Cauchemez S. Reconstructing antibody dynamics to estimate the risk of influenza virus infection. ***Nat Commun.*** *2022 Mar 23;13(1):1557.*
   [IF 16.6 (2022), rank 6/73 Multidisciplinary Sciences]

Refereed Publications (*joint first author, #corresponding author)

1. Xiong W, Huang X, Tran TTN, Nguyen TLT, Peng L, Perera RAPM, Fang VJ, Yang B, Wong SS, Leung NHL, Sullivan SG, Ip DKM, Peiris M, Ross TM, Boni MF, Cowling BJ, **Tsang TK#.** Measuring Population Immunity Against Influenza Using Individual Antibody Titers: An Retrospective Observational Study. ***Accepted by Lancet Infectious Disease***
   [IF 56.3 (2022), rank 1/96 infectious disease]
2. Zhang C, Fang VJ, Chan KH, Leung GM, Ip DKM, Peiris JSM, Cowling BJ, Cauchemez S, **Tsang TK#.** Frequent presymptomatic household transmission of influenza A but not influenza B virus. ***Nature Health*** (2026).
3. **Tsang TK#**, Rojas D, Xu F, Xu Y, Zhu X, Halloran ME, Longini I, Yang Y. Estimating Transmissibility of Zika virus in Colombia in the Presence of Surveillance Bias. ***Nat Commun.*** 2025 May 8;16(1):4299.
   [IF 16.6 (2022), rank 6/73 Multidisciplinary Sciences]
4. Xu Y, Li M, Peng L, Wang C, Wan Q, Zhang C, Xiong W, Yang B, **Tsang TK#.** Comparative Analysis of Flight Volume Effects on COVID-19 and Influenza Transmission Across Variable Control Intensities, 2019-2024. J Infect Dis. 2025 Oct 27:jiaf545
   [IF 6.4 (2022), rank 19/96 Infectious Disease]
5. Peng L, Ainslie KE, Huang X, Cowling BJ, Wu P, **Tsang TK#**. Evaluating the association between COVID-19 transmission and mobility in Omicron outbreaks in China. ***Commun Med (Lond).*** 2025 May 20;5(1):188.
6. Wang C, Yuan HY, Lau EHY, Cowling BJ, Ip DKM, **Tsang TK#.** Impact of Population Immunity and Public Health Measures on the Transmission of Omicron Subvariants BA.2 and BA.5 in Hong Kong. ***J Infect Dis.*** 2025 May 29:jiaf287.
   [IF 6.4 (2022), rank 19/96 Infectious Disease]
7. Chen H, Huang X, Wang C, Cowling BJ, **Tsang TK#.** Estimating the waning effectiveness of COVID-19 vaccines from population-level surveillance data in Hong Kong. ***J Infect Dis.*** 2025 Apr 18:jiaf207.
   [IF 6.4 (2022), rank 19/96 Infectious Disease]
8. Young BR, Ho F, Lin Y, Lau EHY, Wu P, Cowling BJ, **Tsang TK#.** Retrospective assessment of vaccination policy optimizations during an Omicron BA.2 epidemic in Hong Kong: A modelling study. J Infect Public Health. 2025 Sep 12;18(12):102972.

9. Wang C, Peng L, Huang X, **Tsang TK#.** Impact of Vaccination and Public Health Measures on the Severity of SARS-CoV-2 Omicron Infections in China: A Systematic Review and Meta-Regression Analysis. ***Vaccines (Basel).*** 2025 Jul 12;13(7):747
10. Dong S, Sun Y, Ni S, Tian Y, Feng Z, Jia L, Wang X, Zhang D, Wang Q, **Tsang TK#,** Yang P# Interplay Between Viral Shedding, Age, and Symptoms in Individual Infectivity of COVID-19 Breakthrough Infections in Households. ***Vaccines (Basel).*** 2025 Mar 19;13(3):329
11. Young BR, Ho F, Lin Y, Lau EHY, Wu P, Cowling BJ, **Tsang TK**. Optimization of school closures during an Omicron epidemic in Hong Kong: a modelling study. Interface Focus. 2025 Sep 26;15(4):20250016.
12. Yang B, Gostic K, Adam DC, Fox A, Zhang R, Einav T, Peng L, Wong SS, **Tsang TK**, Cummings DAT, Sullivan SG, Cobey S, Cowling BJ. Breadth of influenza A antibody cross-reactivity varies by virus isolation interval and subtype. ***Nat Microbiol.*** 2025. https://doi.org/10.1038/s41564-025-02033-4
13. Liu C, **Tsang TK**, Sullivan SG, Cowling BJ, Yang B. Duration of neutralizing responses and COVID-19 vaccination protection and correlates of protection assessment: A meta-analysis. ***Nat Commun.*** 2025 May 22;16(1):4748.
    [IF 16.6 (2022), rank 6/73 Multidisciplinary Sciences]
14. Chen D, Adam DC, Lau YC, Wang D, Lim WW, Ho F, **Tsang TK**, Lau EHY, Wu P, Wallinga J, Cowling BJ, Ali ST. Investigating setting-specific superspreading potential and generation intervals of COVID-19 in Hong Kong. ***Nat Commun***. 2025 Jul 1;16(1):5816
    [IF 16.6 (2022), rank 6/73 Multidisciplinary Sciences]
15. Poukka E, Murphy C, Mak L, Cheng SMS, Peiris M, **Tsang TK**, Sullivan SG, Cowling BJ. Bias in control selection associated with the use of rapid tests in influenza vaccine effectiveness studies. ***Int J Epidemiol***. 2025 Jun 11;54(4):dyaf089.
16. Meng Y, Lin Y, Xiong W, Lau EHY, Ho F, Wong JY, Wu P, **Tsang TK**, Cowling BJ, Yang B. Effective real-time transmission estimations incorporating population viral load distributions amid SARS-CoV-2 variants and pre-existing immunity. ***J Infect Dis***. 2025 Mar 17;231(3):684-691.
    [IF 6.4 (2022), rank 19/96 Infectious Disease]
17. Chua H, **Tsang TK**, Lee SL, Chan ELY, Kwan MYW, Wong JSC, Peiris M, Sullivan SG, Cowling BJ. Waning in influenza vaccine effectiveness against influenza A(H1N1)pdm09-associated hospitalization in children in 2012/13. ***Epidemiol Infect***. 2025 Nov 24:1-25
    [IF 4.2 (2022), rank 74/207 public, environmental & occupational health]
18. **Tsang TK#**, Du Q, Cowling BJ, Viboud C. An adaptive weight ensemble approach to forecast influenza activity in an irregular seasonality context. ***Nat Commun***. 2024 Oct 4;15(1):8625.
    [IF 16.6 (2022), rank 6/73 Multidisciplinary Sciences]
19. **Tsang TK#**, Sullivan SG, Meng Y, Lai FTT, Fan M, Huang X, Lin Y, Peng L, Zhang C, Yang B, Ainslie KEC, Cowling BJ. Evaluating the impact of extended dosing intervals on mRNA COVID-19 vaccine effectiveness in adolescents. ***BMC Med***. 2024 Sep 12;22(1):384.
    [IF 9.3 (2022), rank 19/168 Medicine, General & Internal]
20. **Tsang TK#**, Sullivan SG, Huang X, Wang C, Wang Y, Nealon J, Yang B, Ainslie KEC, Cowling BJ. Prior infections and effectiveness of SARS-CoV-2 vaccine in test-negative study: A systematic review and meta-analysis. ***Am J Epidemiol.*** *2024 Jun 20:kwae142.*
    [IF 5.0 (2022), rank 48/207 public, environmental & occupational health]
21. **Tsang TK**, Du RQR, Fang VJ, Lau EHY, Chan KH, Chu DKW, Ip DKM, Peiris JSM, Leung GM, Cauchemez S, Cowling BJ. Decreased risk of non-influenza respiratory infection after influenza B virus infection in children. Epidemiol Infect. 2024 Apr 8:1-21.

[IF 4.2 (2022), rank 74/207 public, environmental & occupational health]

22. Zhang C, Fang VJ, Chan KH, Leung GM, Ip DKM, Peiris JSM, Cowling BJ, **Tsang TK#**. Interplay between viral shedding, age, and symptoms on individual infectiousness of influenza cases in households. J Infect Dis. 2024 Aug 28:jiae434.
[IF 6.4 (2022), rank 19/96 Infectious Disease]

23. Peng L, Huang X, Wang C, Xin H, Cowling BJ, Wu P, **Tsang TK#.** Comparative epidemiology of outbreaks caused by SARS-CoV-2 Delta and Omicron variants in China ***Epidemiol Infect.*** *2024 Mar 19;152:e43.*
[IF 4.2 (2022), rank 74/207 public, environmental & occupational health]

24. Young BR, Ho F, Lin Y, Lau EHY, Cowling BJ, Wu P, **Tsang TK#**. Estimation of the Time-Varying Effective Reproductive Number of COVID-19 Based on Multivariate Time Series of Severe Health Outcomes. ***J Infect Dis.*** 2024 Feb 14;229(2):502-506.
[IF 6.4 (2022), rank 19/96 Infectious Disease]

25. Peng H, Cai XQ, Zhen J, Shen JJ, Shen HS, Wu XH, Zhou YB, Ren L, Wu J, Xu A, **Tsang TK#,** Cheung BMY, Li C. The association between circulating fatty acids and stroke in hypertensive patients. Am J Med Sci. 2024 Mar 8:S0002-9629(24)01106-6.
[IF 3.1 (2022), rank 75/169 Medicine, General & Internal]

26. Jia JZ, Cohen CA, Gu H, McLean MR, Varadarajan R, Bhandari N, Peiris M, Leung GM, Poon LLM, **Tsang TK**, Chung AW, Cowling BJ, Leung NHL, Valkenburg SA. Influenza antibody breadth and effector functions are immune correlates from acquisition of pandemic infection of children. Nat Commun. 2024 Apr 13;15(1):3210.

27. **Tsang TK#,** Wang C, Fang VJ, Perera RAPM, So HC, Ip DKM, Leung GM, Peiris JSM, Cauchemez S, Cowling BJ. Reconstructing household transmission dynamics to estimate the infectiousness of asymptomatic influenza virus infections. ***Proc Natl Acad Sci U S A.*** 2023 Aug 15;120(33):e2304750120.
[IF 11.1 (2022), rank 8/73 Multidisciplinary Sciences]

28. **Tsang TK#,** Huang X, Fong MW, Wang C, Lau EHY, Wu P, Cowling BJ. Effects of School-Based Preventive Measures on COVID-19 Incidence, Hong Kong, 2022. ***Emerg Infect Dis.*** 2023 Sep;29(9):1850-1854.
[IF 11.8 (2022), rank 11/96 infectious disease]

29. **Tsang TK#,** Huang X, Wang C, Chen S, Yang B, Cauchemez S, Cowling BJ. The effect of variation of individual infectiousness on SARS-CoV-2 transmission in households. ***eLife***. 2023 Mar 7;12:e82611.
[IF 7.7 (2022), rank 8/92 Biology]

30. **Tsang TK#,** Wang C, Tsang NNY, Fang VJ, Perera RAPM, Malik Peiris JS, Leung GM, Cowling BJ, Ip DKM. Impact of host genetic polymorphisms on response to inactivated influenza vaccine in children. ***NPJ Vaccines.*** 2023 Feb 20;8(1):21.
[IF 9.2 (2022), rank 17/136 Medicine, Research & Experimental]

31. **Tsang TK#**, Huang X, Guo Y, Lau EHY, Cowling BJ, Ip DKM. Monitoring School Absenteeism for Influenza-Like Illness Surveillance: Systematic Review and Meta-analysis. ***JMIR Public Health Surveill***. 2023 Jan 11;9:e41329.
[IF 8.5 (2022), rank 15/207 public, environmental & occupational health]

32. **Tsang TK#**, Gostic KM, Chen S, Wang Y, Arevalo P, Lau EHY, Cobey S, Cowling BJ. Investigation of impact of childhood immune imprinting on birth year-specific risk of clinical infection during influenza A virus epidemics in Hong Kong. ***J Infect Dis***. 2023 Jan 13:jiad009
[IF 6.4 (2022), rank 19/96 Infectious Disease]

33. Xiong W, Cowling BJ, **Tsang TK#.** Influenza Resurgence after Relaxation of Public Health and Social Measures, Hong Kong, 2023 ***Emerg Infect Dis.*** 2023 Oct 26;29(12)
[IF 11.8 (2022), rank 11/96 infectious disease]

34. Sullivan SG, Khvorov A, Huang X, Wang C, Ainslie KEC, Nealon J, Yang B, Cowling BJ, **Tsang TK**. The need for a clinical case definition in test-negative design studies estimating vaccine effectiveness. ***NPJ Vaccines***. 2023 Aug 12;8(1):118.
[IF 9.2 (2022), rank 17/136 Medicine, Research & Experimental]
35. Xiong W, Peng L, **Tsang TK**#, Cowling BJ. Epidemiology of SARS-CoV-2 Omicron BA.5 Infections, Macau, June-July 2022. ***Emerg Infect Dis***. 2023 Feb;29(2):453-456.
[IF 11.8 (2022), rank 11/96 infectious disease]
36. Ho F, Parag KV, Adam DC, Lau EHY, Cowling BJ, **Tsang TK#.** Accounting for the Potential of Overdispersion in Estimation of the Time-varying Reproduction Number. ***Epidemiology***. 2023 Mar 1;34(2):201-205
[IF 5.4 (2022), rank 41/207 public, environmental & occupational health]
37. Lin Y, Wu P, **Tsang TK**, Wong JY, Lau EHY, Yang B, Leung GM, Cowling BJ. Viral kinetics of SARS-CoV-2 following onset of COVID-19 in symptomatic patients infected with the ancestral strain and omicron BA.2 in Hong Kong: a retrospective observational study. ***Lancet Microbe***. 2023 Sep;4(9):e722-e731.
[IF 38.2 (2022), rank 2/96 Infectious Disease]
38. Xie R, Edwards KM, Adam DC, Leung KSM, **Tsang TK**, Gurung S, Xiong W, Wei X, Ng DYM, Liu GYZ, Krishnan P, Chang LDJ, Cheng SMS, Gu H, Siu GKH, Wu JT, Leung GM, Peiris M, Cowling BJ, Poon LLM, Dhanasekaran V. Resurgence of Omicron BA.2 in SARS-CoV-2 infection-naive Hong Kong. ***Nat Commun.*** *2023 Apr 27;14(1):2422*
[IF 16.6 (2022), rank 6/73 Multidisciplinary Sciences]
39. Yang B, Lin Y, Xiong W, Liu C, Gao H, Ho F, Zhou J, Zhang R, Wong JY, Cheung JK, Lau EHY, **Tsang TK**, Xiao J, Wong IOL, Martín-Sánchez M, Leung GM, Cowling BJ, Wu P. Comparison of control and transmission of COVID-19 across epidemic waves in Hong Kong: an observational study. ***Lancet Reg Health West Pac.*** 2023 Nov 22;43:100969.
[IF 7.1 (2022), rank 26/207 public, environmental & occupational health]
40. Lun P, Ning K, Wang Y, Ma TSW, Flores FP, Xiao X, Subramaniam M, Abdin E, Tian L, **Tsang TK**, Leung K, Wu JT, Cowling BJ, Leung GM, Ni MY. COVID-19 Vaccination Willingness and Reasons for Vaccine Refusal. ***JAMA Netw Open***. 2023 Oct 2;6(10):e2337909.
[IF 13.8 (2022), rank 15/168 Medicine, General & Internal]
41. Du Z, Zhang X, Wang L, Yao S, Bai Y, Tan Q, Xu X, Pei S, Xiao J, **Tsang TK**, Liao Q, Lau EHY, Wu P, Gao C, Cowling BJ. Characterizing Human Collective Behaviors During COVID-19 - Hong Kong SAR, China, 2020. ***China CDC Wkly***. 2023 Jan 27;5(4):71-75
[IF 4.7 (2022), rank 58/207 public, environmental & occupational health]
42. **Tsang TK**, Perera RAPM, Fang VJ, Wong JY, Shiu EY, So HC, Ip DKM, Malik Peiris JS, Leung GM, Cowling BJ, Cauchemez S. Reconstructing antibody dynamics to estimate the risk of influenza virus infection. ***Nat Commun.*** *2022 Mar 23;13(1):1557.*
[IF 16.6 (2022), rank 6/73 Multidisciplinary Sciences]
43. **Tsang TK**, Lam KT, Liu Y, Fang VJ, Mu X, Leung NHL, Peiris JSM, Leung GM, Cowling BJ, Tu W. Investigation of CD4 and CD8 T cell-mediated protection against influenza A virus in a cohort study ***BMC Med***. 2022 Jul 21;20(1):230.
[IF 9.3 (2022), rank 19/168 Medicine, General & Internal]
44. **Tsang TK#,** Wang C, Fang VJ, Perera RAPM, So HC, Ip DKM, Peiris JSM, Leung GM, Cauchemez S, Cowling BJ. Indirect Protection from Vaccinating Children against Influenza A Virus Infection in Households. ***Viruses***. 2022 Sep 21;14(10):2097.
[IF 4.7 (2022), rank 15/36 Virology]
45. **Tsang TK***, Fang LQ*, Zhang A, Jiang FC, Ruan SM, Liu LZ, Cowling BJ, Liu W, Yang Y. Variability in transmission risk of SARS-CoV-2 in close contact settings: A contact tracing study in Shandong Province, China ***Epidemics.*** *2022 Mar 9;39:100553*

[IF 3.8 (2022), rank 50/96 Infectious Disease]

46. Xiong W, **Tsang TK**#, Perera RAPM, Leung NHL, Fang VJ, Barr IG, Peiris JSM, Cowling BJ. Biphasic waning of hemagglutination inhibition antibody titers after influenza vaccination in children. ***J Infect Dis.*** *2022 Apr 5:jiac117.*
[IF 6.4 (2022), rank 19/96 Infectious Disease]

47. Ho F, **Tsang TK**#, Gao H, Xiao J, Lau EHY, Wong JY, Wu P, Leung GM, Cowling BJ. Restaurant-Based Measures to Control Community Transmission of COVID-19, Hong Kong ***Emerg Infect Dis****. 2022 Mar;28(3):759-761.*
[IF 11.8 (2022), rank 11/96 infectious disease]

48. Wang C, Huang X, Lau EHY, Cowling BJ, **Tsang TK**#. Association Between Population-Level Factors and Household Secondary Attack Rate of SARS-CoV-2: A Systematic Review and Meta-analysis. ***Open Forum Infect Dis.*** 2022 Dec 16;10(1):ofac676.
[IF 4.2 (2022), rank 44/96 Infectious Disease]

49. Yang B, **Tsang TK**, Gao H, Lau EHY, Lin Y, Ho F, Xiao J, Wong JY, Adam DC, Liao Q, Wu P, Cowling BJ, Leung GM. Universal Community Nucleic Acid Testing for Coronavirus Disease 2019 (COVID-19) in Hong Kong Reveals Insights Into Transmission Dynamics: A Cross-Sectional and Modeling Study. ***Clin Infect Dis.*** 2022 Aug 24;75(1):e216-e223.
[IF 11.8 (2022), rank 10/96 Infectious Disease]

50. Cheung JTL, **Tsang TK**, Yen HL, Perera RAPM, Mok CKP, Lin YP, Cowling BJ, Peiris M. Determining Existing Human Population Immunity as Part of Assessing Influenza Pandemic Risk ***Emerg Infect Dis****. 2022 May;28(5):977-985.*
[IF 11.8 (2022), rank 11/96 infectious disease]

51. Yang B, Huang X, Gao H, Leung NH, **Tsang TK**, Cowling BJ. Immunogenicity, efficacy, and safety of SARS-CoV-2 vaccine dose fractionation: a systematic review and meta-analysis. ***BMC Med.*** 2022 Oct 25;20(1):409.
[IF 9.3 (2022), rank 19/168 Medicine, General & Internal]

52. Yang B, Sullivan SG, Du Z, Tsang TK, Cowling BJ. Effectiveness of International Travel Controls for Delaying Local Outbreaks of COVID-19. ***Emerg Infect Dis.*** 2022 Jan;28(1):251-253.
[IF 11.8 (2022), rank 11/96 infectious disease]

53. Chen D, Lau YC, Xu XK, Wang L, Du Z, **Tsang TK**, Wu P, Lau EHY, Wallinga J, Cowling BJ, Ali ST. Inferring time-varying generation time, serial interval, and incubation period distributions for COVID-19. ***Nat Commun***. 2022 Dec 13;13(1):7727.
[IF 16.6 (2022), rank 6/73 Multidisciplinary Sciences]

54. Du Z, Wang L, Shan S, Lam D, **Tsang TK,** Xiao J, Gao H, Yang B, Ali ST, Pei S, Fung IC, Lau EHY, Liao Q, Wu P, Meyers LA, Leung GM, Cowling BJ. Pandemic fatigue impedes mitigation of COVID-19 in Hong Kong***. Proc Natl Acad Sci U S A.*** 2022 Nov 29;119(48):e2213313119.
[IF 11.1 (2022), rank 8/73 Multidisciplinary Sciences]

55. Ali ST, Lau YC, Shan S, Ryu S, Du Z, Wang L, Xu XK, Chen D, Xiong J, Tae J, **Tsang TK**, Wu P, Lau EHY, Cowling BJ. Prediction of upcoming global infection burden of influenza seasons after relaxation of public health and social measures during the COVID-19 pandemic: a modelling study. ***Lancet Glob Health***. 2022 Nov;10(11):e1612-e1622.
[IF 34.3 (2022), rank 2/207 public, environmental & occupational health]

56. Lin Y, Yang B, Cobey S, Lau EHY, Adam DC, Wong JY, Bond HS, Cheung JK, Ho F, Gao H, Ali ST, Leung NHL, **Tsang TK**, Wu P, Leung GM, Cowling BJ. Incorporating temporal distribution of population-level viral load enables real-time estimation of COVID-19 transmission ***Nat Commun.*** *2022 Mar 3;13(1):1155.*
[IF 16.6 (2022), rank 6/73 Multidisciplinary Sciences]

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[IF 6.4 (2022), rank 19/96 Infectious Disease]
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[IF 13.8 (2022), rank 15/168 Medicine, General & Internal]
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[IF 7.1 (2022), rank 26/207 public, environmental & occupational health]
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China: a modelling study. ***Lancet Public Health.*** *2020 May;5(5):e289-e296.* [IF 50.0 (2022), rank 1/207 public, environmental & occupational health]

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79. **Tsang TK**, Lau LL, Cauchemez S, Cowling BJ. Review of household transmission of influenza virus. ***Trends Microbiol.*** *2016;24(2):123-33* [IF 15.9 (2022), rank 6/135 Mircobiology]

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[IF 5.4 (2022), rank 41/207 public, environmental & occupational health]
81. **Tsang TK**, Cauchemez S, Fang VJ, Chan KH, Ip DK, Leung GM, Peiris JS, Cowling BJ. Individual correlates of infectivity of influenza virus infections in households ***PLoS One.*** 2016;11(5):e0154418.
[IF 3.7 (2022), rank 26/73 Multidisciplinary Sciences]
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[IF 4.2 (2022), rank 74/207 public, environmental & occupational health]
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[IF 6.4 (2022), rank 19/96 Infectious Disease]
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[IF 11.8 (2022), rank 10/96 Infectious Disease]
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[IF 6.4 (2022), rank 19/96 Infectious Disease]
89. Virlogeux V, Li M, **Tsang TK**, Feng L, Fang VJ, Jiang H, Wu P, Zheng J, Lau EH, Cao Y, Qin Y, Liao Q, Yu H, Cowling BJ. Estimating the Distribution of the Incubation Periods of Human Avian Influenza A(H7N9) Virus Infections. ***Am J Epidemiol***. *2015;182(8):723-9.*
[IF 5.0 (2022), rank 48/207 public, environmental & occupational health]
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[IF 5.0 (2022), rank 48/207 public, environmental & occupational health]

91. **Tsang TK**, Cauchemez S, Perera RA, Freeman G, Fang VJ, Ip DK, Leung GM, Peiris JS, Cowling BJ. Association between antibody titers and protection against influenza virus infection within households. ***J Infect Dis.*** 2014;210(5):684-92.
[IF 6.4 (2022), rank 19/96 Infectious Disease]
92. Wang C*, Yu H*, Horby P*, Cao B*, Wu P*, Yang S*, Gao H*, Li H*, **Tsang TK***, Liao Q, Gao Z, Ip DK, Jia H, Jiang H, Liu B, Ni MY, Dai X, Liu F, Kinh NV, Liem NT, Hien TT, Li Y, Yang J, Wu JT, Zheng Y, Leung GM, Farrar J, Cowling BJ, Uyeki TM, Li L. Comparison of patients hospitalized with influenza A H7N9, H5N1, and 2009 pandemic H1N1. ***Clin Infect Dis.*** 2014 ;58(8):1095-103.
[IF 11.8 (2022), rank 10/96 Infectious Disease]
93. Lau EH, Zheng J, **Tsang TK**, Liao Q, Lewis B, Brownstein JS, Sanders S, Wong JY, Mekaru SR, Rivers C, Wu P, Jiang H, Li Y, Yu J, Zhang Q, Chang Z, Liu F, Peng Z, Leung GM, Feng L, Cowling BJ, Yu H. Accuracy of epidemiological inferences based on publicly available information: retrospective comparative analysis of line lists of human cases infected with influenza A(H7N9) in China. ***BMC Med.*** 2014 May 28;12:88.
[IF 9.3 (2022), rank 19/168 Medicine, General & Internal]
94. Liao Q, Ip DK, **Tsang TK**, Cao B, Jiang H, Liu F, Zheng J, Peng Z, Wu P, Huai Y, Lau EH, Feng L, Leung GM, Yu H, Cowling BJ. A clinical prediction rule for diagnosing human infections with avian influenza A(H7N9) in a hospital emergency department setting. ***BMC Med.*** 2014 Aug 5;12:127.
[IF 9.3 (2022), rank 19/168 Medicine, General & Internal]
95. Wu P, Jiang H, Wu JT, Chen E, He J, Zhou H, Wei L, Yang J, Yang B, Qin Y, Fang VJ, Li M, **Tsang TK,** Zheng J, Lau EH, Cao Y, Chai C, Zhong H, Li Z, Leung GM, Feng L, Gao GF, Cowling BJ, Yu H. Poultry market closures and human infection with influenza A(H7N9) virus, China, 2013-14. ***Emerg Infect Dis.*** 2014;20(11):1891-4.
[IF 11.8 (2022), rank 11/96 infectious disease]
96. Feng L, Wu JT, Liu X, Yang P, **Tsang TK**, Jiang H, Wu P, Yang J, Fang VJ, Qin Y, Lau EH, Li M, Zheng J, Peng Z, Xie Y, Wang Q, Li Z, Leung GM, Gao GF, Yu H, Cowling BJ. Clinical severity of human infections with avian influenza A(H7N9) virus, China, 2013/14. ***Euro Surveill.*** 2014;19(49). pii: 20984.
[IF 19.0 (2022), rank 5/96 infectious disease]
97. Yu H, Cowling BJ, Feng L, Lau EH, Liao Q, **Tsang TK**, Peng Z, Wu P, Liu F, Fang VJ, Zhang H, Li M, Zeng L, Xu Z, Li Z, Luo H, Li Q, Feng Z, Cao B, Yang W, Wu JT, Wang Y, Leung GM. Human infection with avian influenza A H7N9 virus: an assessment of clinical severity**.** ***Lancet.*** 2013 Jul 13;382(9887):138-45.
[IF 168.9 (2022), rank 1/168 Medicine, General & Internal]
98. Cowling BJ, Jin L, Lau EH, Liao Q, Wu P, Jiang H, **Tsang TK**, Zheng J, Fang VJ, Chang Z, Ni MY, Zhang Q, Ip DK, Yu J, Li Y, Wang L, Tu W, Meng L, Wu JT, Luo H, Li Q, Shu Y, Li Z, Feng Z, Yang W, Wang Y, Leung GM, Yu H. Comparative epidemiology of human infections with avian influenza A H7N9 and H5N1 viruses in China: a population-based study of laboratory-confirmed cases. ***Lancet.*** 2013 Jul 13;382(9887):129-37.
[IF 168.9 (2022), rank 1/168 Medicine, General & Internal]

Manuscripts under review / in revision

99. **Tsang TK**#, Sullivan SG, Huang X, Wang C, Peng L, Yang B, Cowling BJ. Intensity of public health and social measures is associated with SARS-CoV-2 vaccine effectiveness estimates in test-negative studies. ***Under second round external review at Nature Health***
100. Peng L, Guo Y, Tsang NNY, Huang X, So HC, Cowling BJ, Ip DKM, **Tsang TK#.** Multi-Source Agent-Based Modeling to Optimize Influenza Mitigation Strategies in Hong Kong. ***Revise and Resubmit at Nature Communications***

101. **Tsang TK**#, Zhang C, Camaglia F, Monto AS, Cauchemez S, Cowling BJ#. Infectious Viral Titers Serve as Biomarkers for Influenza Transmission Risk in Household Settings. ***Under external review at Lancet Microbe***

Presentations
Invited Talks

1. Research seminar in CEIRR Computational Modeling Core, Webinars, Feb 2026
2. Symposium on Bayesian Methods in Health Research in The University of Hong Kong, Hong Kong, China, June 2025
3. Beijing Center for Disease Prevention and Control, Beijing, China, Oct 2024
4. Statistical advances in surveillance, analysis and forecasting of emerging infectious diseases, Joint Statistics Meeting, Portland, OR, USA, Aug 2024
5. Infectious Disease Control Training Centre, Infection Control Branch, Centre of Health Protection, Hong Kong, Webinars, Feb 2022
6. Department of Biostatistics, Epidemiology and Environmental Health Sciences, Jiann-Ping Hsu College of Public Health, Georgia Southern University, Webinars, Mar 2021
7. Department of Agricultural and biological engineering, University of Florida, Gainesville, FL, USA, Nov 2018
8. NYICE Seminar Series in University of Rochester, Rochester, NY, USA, May 2017
9. Center for inference and dynamics of infectious disease, Seattle, WA, USA, Apr 2016

Contributed Talks

1. Epidemics 10, San Diego, CA, USA, December 2025
2. The tenth ESWI influenza conference, Spain, Oct 2025
3. The fourth Correlates of Protection for Next Generation Influenza Vaccines, Austria, Oct 2025
4. The ninth ESWI influenza conference, Spain, Sep 2023
5. The seventh ESWI influenza conference, virtual conference, Dec 2020
6. Annual Scientific Meeting, Hong Kong College of Community Medicine, Hong Kong, Sep 2015

Contributed Posters

1. Option XII, Brisbane, Australia, October 2024
2. Epidemics 9, Bologna, Italy, December 2023
3. Individual and Population Immunity to Respiratory Viruses, Hong Kong, Nov 2023
4. ISIRV-WHO Virtual Conference, virtual conference, Oct 2021
5. The seventh ESWI influenza conference, virtual conference, Dec 2020
6. Annual Scientific Meeting, Hong Kong College of Community Medicine, Hong Kong, Sep 2020
7. Seventh International Conference on Infectious Disease. Charleston, SC, USA, Dec 2019
8. EPI Research Day, University of Florida, FL, USA, Feb 2019
9. MIDAS Network Meeting, Bethesda, MD, USA, Apr 2018
10. EPI Research Day, University of Florida, FL, USA, Feb 2018
11. ASTMH Annual Meeting, Baltimore, MD, USA, Nov 2017
12. MIDAS Network Meeting, Atlanta, GA, USA, May 2017
13. EPI Research Day, University of Florida, FL, USA, Feb 2017
14. Fifth International Conference on Infectious Disease. Tampa, FL, USA, Dec 2015
15. East-West Alliance Global Symposia, Hong Kong, Oct 2014

16. Annual Scientific Meeting, Hong Kong College of Community Medicine, Hong Kong, Sep 2013

Teaching Experience

1. Course coordinator (Biostatistics) for Introduction to Epidemiology and Biostatistics (GHAD1004), School of Public Health, The University of Hong Kong, Fall 2023-2025
2. Course coordinator for Advanced statistical methods II (CMED 6040), School of Public Health, The University of Hong Kong, Spring 2020-2025
3. Guest Lecturer for Advanced statistical methods I (CMED 6020), School of Public Health, The University of Hong Kong, March 2024
4. Guest Lecturer for Analytic methods for infectious disease, Department of Biostatistics, University of Florida, April 2018
5. Tutor for graduate courses, School of Public Health, The University of Hong Kong, Fall 2013 to Fall 2016
   - Introduction of biostatistics, Fall 2013, Fall 2014, Fall 2015, Fall 2016, Fall 2019
   - Advanced statistical methods I, Spring 2014, Spring 2015, Spring 2016
6. Guest Lecturer for Introduction to statistical software, Influenza Disease Burden Training Workshop, China CDC, Sep 2013
7. Teaching Assistant for undergraduate courses, Department of Mathematics, Hong Kong University of Science of Technology, Fall 2010 to Spring 2012
   - Statistical inference, Fall 2011
   - Time series analysis, Spring 2011, Spring 2012
   - Data analysis, Fall 2010

Research Postgraduate Student Supervision

1. Yani Wu (***HKPFS awardee***, PhD supervisor), Epidemiology and Biostatistics, 2025 - present
2. Haoling Chen (***HKUPS awardee***, PhD supervisor), Infectious Disease Epidemiology and Modelling, 2025 – present
3. Yue Sui (PhD supervisor), Biostatistics and Infectious Disease Modeling, 2025 – present
4. Xiaotong Huang (mphil co-supervisor), Biostatistics and Infectious Disease Modeling, 2025 - present
5. Chengyao Zhang (***HKPFS awardee***, PhD supervisor), Infectious Disease Epidemiology and Modelling, 2023 - present
6. Yanfang Xu (PhD supervisor), Epidemiology and Biostatistics, 2023- present
7. Liping Peng (PhD supervisor), Infectious disease epidemiology and modeling, 2023 – present
8. Yu Meng (PhD supervisor), Infectious disease epidemiology, 2023- present
9. Chang Liu (PhD co-supervisor), Biostatistics and Infectious Disease Modeling, 2023 - present
10. Can Wang (PhD supervisor), Infectious disease epidemiology and modeling, 2022 - present
11. Weijia Xiong (PhD supervisor), Infectious disease epidemiology, 2021 – 2025
12. Benjamin Ross Young (PhD co-supervisor), Infectious disease epidemiology, 2021 – 2025
13. Faith Ho (PhD co-supervisor), Influenza vaccination strategies in older adults, 2019 – 2023

Computational Skills

Languages: R, C++, Python, MATLAB
Statistical Software: SPSS, SAS

Knowledge exchange

1. COVID-19 dashboard from March 2020 to May 2023. Provided daily update on the Rt estimates of COVID-19.
2. R package 'hhdynamics'. Implementation of household transmission models (Tsang et al. Trends in Microbiology 2016).
3. R package 'seroreconstruct'. Implementation of serology modeling (Tsang et al. Nat Commum 2022)
4. R package 'ImmuPop'. Calculation of Population immunity of influenza (Xiong et al.)

Service
Editorial board members
Scientific Reports, PLoS One, BMC Infectious Diseases, AJE Advances: Research in Epidemiology

Statistical Advisory Board
Nature Medicine

Referee
Epidemiology: Nature Medicine, Nature Microbiology, Nature Human Behaviors, Nature Communications, Clinical Infectious Disease, eClinicalMedicine, elife, Emerging Infectious Diseases, Eurosurveillance, American journal of Epidemiology, Epidemiology, Epidemics, PLoS Computational Biology, The Lancet Regional Health - Western Pacific, JMIR Public Health and Surveillance, Influenza and other Respiratory Viruses, Scientific Report, PLoS One, International Journal of infectious disease, BMJ open, PLoS neglected tropical diseases, Vaccine, Epidemiology and infection, Cell Reports Medicine, Reviews in Medical Virology, Hepatology Communications, Diabetes Obesity and Metabolism, BMC Infectious Diseases, BMC Health Services Research
Statistics: Statistics in Medicine, Journal of the Royal Statistical Society: Series A, Statistical in Bioscience

Examiner of PhD degree:

1. Mr Yang Li, PhD Candidate in HKU, 2025
2. Miss Yawei Wang, PhD Candidate in HKU, 2025
3. Miss Shuyi Zhong, PhD Candidate in HKU, 2025
4. Mr NG Chun Man, PhD Candidate in HKU, 2023
5. Dr TAM Greta Chun-huen, PhD Candidate in HKU, 2023
6. Miss FONG Min Whui, PhD Candidate in HKU, 2022
7. Miss Valentina Costantino, PhD Candidate in University of New South Wales, 2021

Honors and Awards

1. Highly Cited Researcher (ESI), Clarivate Essential Science Indicators -- Top 1% by citations in Cross-Field, 2022, 2023, 2024 and 2025
2. Faculty Outstanding Research Output Award for Adam et al. 2020 Nature Medicine paper, The University of Hong Kong, 2022
3. Faculty Outstanding Research Output Award for Tsang et al. 2019 Nature Communications paper, The University of Hong Kong, 2020
4. Faculty Outstanding Research Output Award for Cowling et al. 2013 Lancet paper and Yu et al. 2013 Lancet paper, The University of Hong Kong, 2014

5. L'Oreal Scholarship for 3 months visit to work with Dr Simon Cauchemez in Mathematical Modelling of Infectious Diseases, Institut Pasteur, 2014
6. Postgraduate studentship award, Hong Kong University of Science of Technology, 2010-2012
7. Swire University Scholarship, Hong Kong University of Science of Technology, 2008-2010
8. Dean's list Award, Hong Kong University of Science of Technology, 2008-2010